Type MS · Explain homeostasis and negative feedback

Answer 1 of 6 · Question 4 part c · 4 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

4 (a)	• maintain a constant/stable/optimum; • internal environment; • despite changes in the external environment;		3
(b) (i)	B; (glucagon) *A is secreted by the kidneys* *C is insulin and reduces blood glucose* *D is testosterone and it is secreted by the testes .*		1
(ii)	C; (insulin) *A is secreted by the kidneys* *C is glucagon and increases blood glucose* *D is testosterone and it is secreted by the testes.*		1
(iii)	P = glycogen; Q = glucose; R = fat/lipid; S = respiration;		4
(c)	• low levels of blood glucose; • reduce insulin release; • until balance/correct/normal level is reached/increases blood glucose; • secretion of insulin starts again/glucagon stopped;	accept reverse argument for glucagon	4

Total : 14 marks

Type MS · Explain homeostasis and negative feedback

Answer 1 of 6 · Question 4 part c · 4 marks

2022 JAN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain homeostasis and negative feedback

Answer 2 of 6 · Question 7 part c · 2 marks

2020 NOV · Paper 01 (regional R) · find part c in this mark scheme extract

7 (a)	Same diagram with the following modifications: • shunt vessel noticeably more narrow(1) • capillaries noticeably wider(1)		2
(b)	Any four from: • capillaries dilated/get wider/vasodilation(1) • to allow greater blood flow nearer to skin surface(1) • shunt vessel constricts/becomes more narrow(1) • to divert more blood through capillaries(1) • more heat lost through skin's surface/by radiation(1)	Allow blood vessels for capillaries Ignore arteries/veins	4
(c)	• (change in body temperature) detected by the hypothalamus(1) • impulses sent to effector/named effector (1) • temperature change reversed/named reverse process (1)	Allow reverse argument for both marking points Allow signals for impulses Allow temperature returns to normal	2

Total for Question 7 = 8 Marks

Type MS · Explain homeostasis and negative feedback

Answer 3 of 6 · Question 4 part a · 3 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

4 (a)	• maintain a constant/stable/optimum; • internal environment; • despite changes in the external environment;		3
(b) (i)	B; (glucagon) *A is secreted by the kidneys* *C is insulin and reduces blood glucose* *D is testosterone and it is secreted by the testes .*		1
(ii)	C; (insulin) *A is secreted by the kidneys* *C is glucagon and increases blood glucose* *D is testosterone and it is secreted by the testes.*		1
(iii)	P = glycogen; Q = glucose; R = fat/lipid; S = respiration;		4
(c)	• low levels of blood glucose; • reduce insulin release; • until balance/correct/normal level is reached/increases blood glucose; • secretion of insulin starts again/glucagon stopped;	accept reverse argument for glucagon	4

Total : 14 marks

Type MS · Explain homeostasis and negative feedback

Answer 3 of 6 · Question 4 part a · 3 marks

2022 JAN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain homeostasis and negative feedback

Answer 4 of 6 · Question 3 part d · 3 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

Number				
3	(a)	any two from • protection/named example/barrier against pathogens; • sensory reception/named example; • excretion		2
	(b)	• looks red; • vasodilation/more blood in skin capillaries/blood flows closer to skins surface; • allows more heat loss/heat energy released; • to lower body temperature;		4
	(c) (i)	• as temperature increases, rate of sweating increases; • up until 34°C/ between 32 °C - 35.9 °C, little/no change; • increases, rapidly/significantly from 35.9 °C - 38 °C;	Allow any temperature between 34°C and 35.9 °C for 35.9 °C e.g. between 32°C and 34 °C	2
	(ii)	• absorb sweat with cotton wool/absorbent paper; • fixed area of skin/under arm pits; • reference to measuring mass of absorbent material before and after; or • measure time; • for cobalt chloride paper to change colour;		2
	(d)	• enzymes affected/to maintain enzyme activity/ref to body temperature being optimum for enzymes; • too high a temperature denatured/prevent denaturation; • too low, inactive/less active/prevent enzymes becoming inactive;		3

Total 13 marks

Type MS · Explain homeostasis and negative feedback

Answer 4 of 6 · Question 3 part d · 3 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain homeostasis and negative feedback

Answer 5 of 6 · Question 8 part b.iii · 2 marks

2024 JUN · Paper 02 (regional R) · find part b.iii in this mark scheme extract

8 (a)	any three from • released from pancreas; • when glucose levels rise; • causes/stimulates conversion of glucose to glycogen; • in liver/muscles;		(3)
(b) (i)	• vasodilation; • more blood flows at/near the surface of the skin; • heat (from blood) radiated (away from body)/increased heat loss; • hairs lie flat; • less air trapped; • (more) heat radiated;		(3)
(ii)	vasoconstriction/blood vessels in the skin narrowed shivering/increased metabolic rate/hairs stand on end;		(1)
(iii)	• hypothalamus detects rise/fall/change in body temperature; • (named) mechanism reverses the change/brings back to set point/normal;	e.g. vasodilation, increase in sweating	(2)
(c)	• enzymes lose kinetic energy; • enzyme activity decreases/works slowly; • less enzyme substrate complexes formed/ less collisions between enzyme and substrate/binding with substrate; • rate of chemical reactions/metabolism slow down; • reference to symptom of hypothermia e.g. death/weak pulse/slower heart/breathing rate;		(5)

Total for Question 8 = 14 marks

Type MS · Explain homeostasis and negative feedback

Answer 6 of 6 · Question 3 part a · 4 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

3 (a)	• control of/maintenance of • constant internal environment; • despite changes in external environment; • ref negative feedback;	ignore specific examples	4
(b)	box 1 hypothalamus; detects rise/increase in blood temperature; box 2 arterioles dilate/vasodilation; box 3 sweat evaporates; box 4 (more) heat loss (from skin); by radiation;	ignore thermoregulatory centre	6
(c)	• enzymes; • denatured; • fewer/no enzyme substrate complexes formed/substrate can't bind to active site/reduced enzyme activity/reaction; • reduced metabolic activity/rate of reactions in body reduced;		4
			14 marks

Type MS · Explain homeostasis and negative feedback

Answer 6 of 6 · Question 3 part a · 4 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks